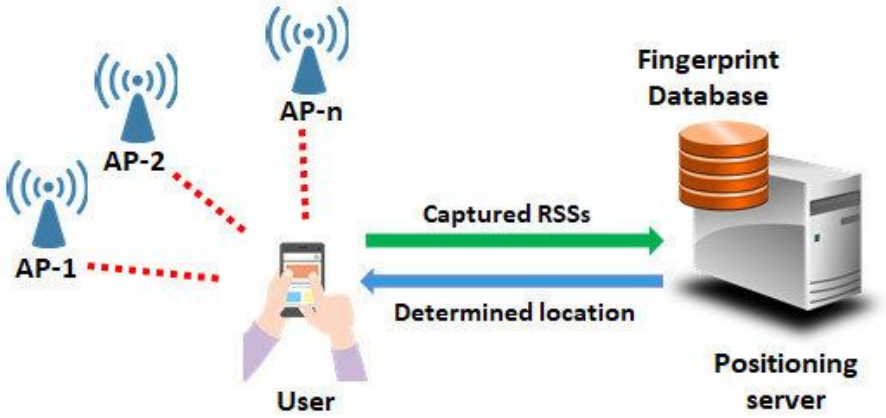
Wifi based Indoor Positioning System

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What is Wifi Positioning System

Wi-Fi positioning system (**WPS**, also abbreviated as **WiPS** or **WFPS**) is a geolocation system that uses the characteristics of nearby Wi-Fi hotspots and other wireless access points to discover where a device is located.

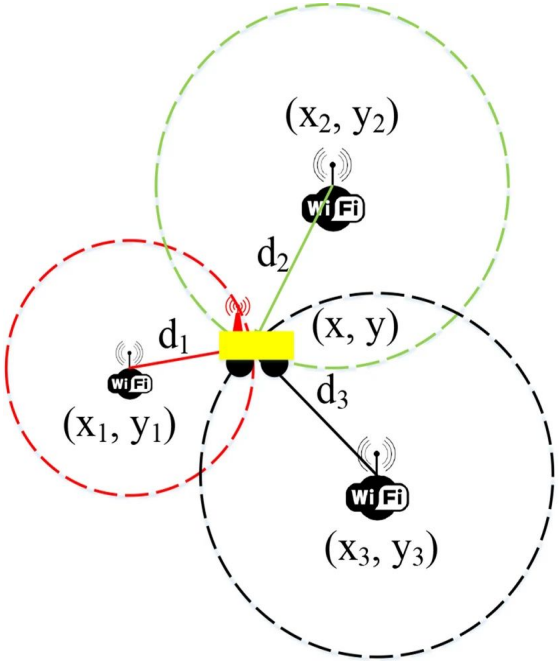
It is used where satellite navigation such as GPS is inadequate due to various causes including multipath and signal blockage indoors, or where acquiring a satellite fix would take too long. Such systems include assisted GPS, urban positioning services through hotspot databases, and indoor positioning systems. Wi-Fi positioning takes advantage of the rapid growth in the early 21st century of wireless access points in urban areas



Wi-Fi indoor positioning techniques are classified into 2 categories:

	Signal propagation mode	Location fingerprinting method
Technique	RSSI, ToA, TDoA, AoA, AoD	Fingerprinting
Measurements	Signal strength, angles	Fingerprint database
Accuracy	Medium	High
Time cost	Low	High
Distance	Low	Medium
Algorithm	Deterministic	Deterministic and probabilistic
Specification	Time based	Range based
Advantages	Additional special-purpose hardware is not required Acceptable cost and accessibility	Addresses some of the problems related to non-line-of-sight and multipath propagation Does not require exact locations of access points]
Disadvantages	More difficult regarding synchronization of receiver/transmitter-side times and accuracy of short-range flight time measurements For angle calculation, complex hardware is required to determine the angle of arrival Poor positioning accuracy in real-world indoor environments because of a complex accurate propagation model Multipath is an issue with signal propagation	Time-consuming and laborious Accurate fingerprint data depends on the autonomic construction of a fingerprint database When real-world indoor environments change significantly, the fingerprint database has to be rebuilt

Wi-Fi survey-based indoor positioning techniques are divided into 2 types: Trilateration and Triangulation algorithms



$$\begin{cases}(x-x_1)^2+(y-y_1)^2=d_1^2\\(x-x_2)^2+(y-y_2)^2=d_2^2\\(x-x_3)^2+(y-y_3)^2=d_3^2\end{cases}$$

$$QX=b$$

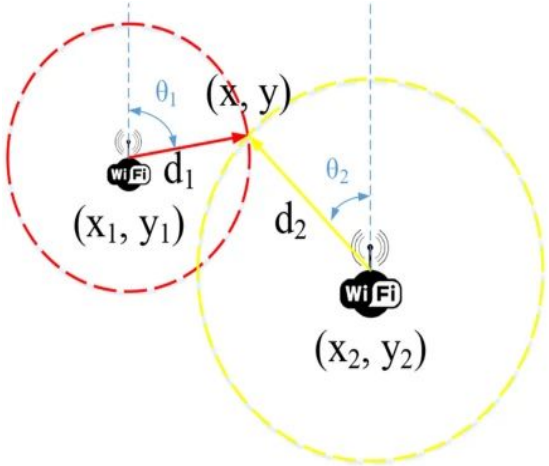
$$X=\begin{bmatrix}x\\y\end{bmatrix}$$

$$Q=\begin{bmatrix}2(x_1-x_2) & 2(y_1-y_2)\\2(x_1-x_3) & 2(y_1-y_3)\end{bmatrix}$$

$$b=\begin{bmatrix}x_1^2-x_2^2+y_1^2-y_2^2+d_2^2-d_1^2\\x_1^2-x_3^2+y_1^2-y_3^2+d_3^2-d_1^2\end{bmatrix}$$

$$X=Q^{-1}b$$

Positioning based on trilateration algorithms



$$\begin{cases}x=d_1\sin(\theta_1)+x_1\\y=d_1\cos(\theta_1)+y_1\\x=-d_2\sin(\theta_2)+x_2\\y=d_2\cos(\theta_2)+y_2\end{cases}$$

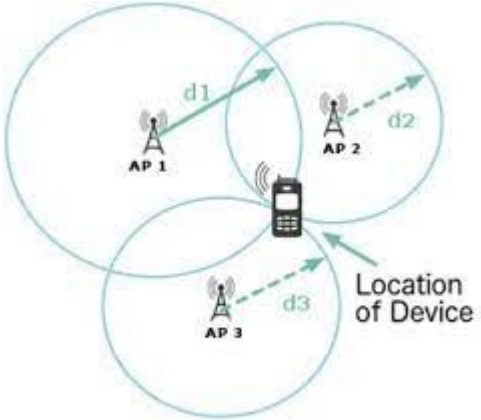
Positioning based on triangulation algorithms

Problem

The problem of Wi-Fi based indoor localization of a device consists in determining the position of client devices with respect to access points. Many techniques exist to accomplish this, and these may be classified into four main types: *received signal strength indication* (RSSI), fingerprinting, angle of arrival (AoA) and time of flight (ToF) based techniques.

In most cases the first step to determine a device's position is to determine the distance between the target client device and a few access points. With the known distances between the target device and access points, trilateration algorithms may be used to determine the relative position of the target device, using the known position of access points as a reference. Alternatively, the angle of arriving signals at a target client device can be employed to determine the device's location based on triangulation algorithms.

A combination of these techniques may be used to increase the accuracy of the system.



Methodology

In RSSI-based localization systems, the accuracy of the final location largely depends on the range. Therefore, in order to obtain higher accuracy, we need to try various possible methods to improve the range accuracy. In equation (1) we get the relationship of RSSI and distance, where the parameter n means the exponential attenuation factor to distance. In the indoor environment, the external factors deeply affect range value; even in the same location of a node, the value of n may be variable with different circumstances. In order to maintain a certain location accuracy, adjusting the value of n is necessary. Here a method is designed to adjust the range value between blind node and beacon, which depends on RSSI and real distance of beacons.

$$RSSI = A - 10nlgd \quad (1)$$

Objectives

- Localization is utilized in various fields to encourage certain activities e.g security, service provision.

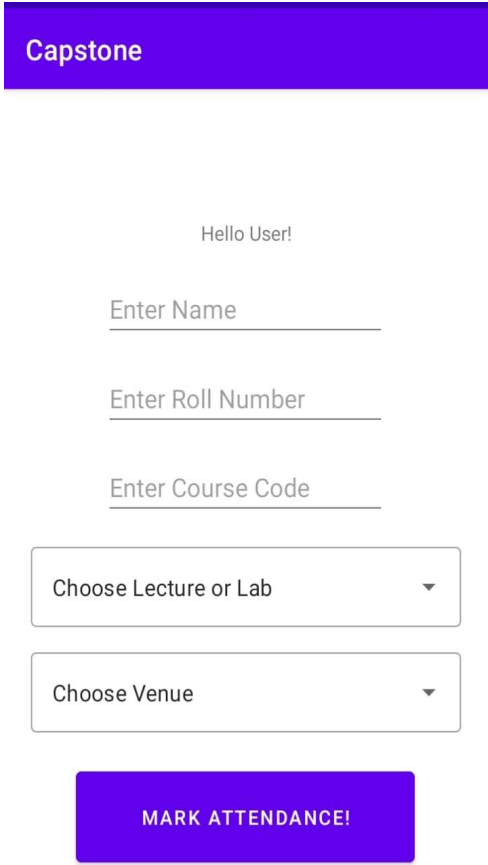
1. Accuracy: Correct and precise location classification.
2. Security: Major and significant difficulties of IoT is security which may be overcome by some secure storage techniques like storing the information on cloud networks.
3. Range: In RSSI-based localization systems, the accuracy of the final location largely depends on the range.

Languages Used : Java, Python, JavaScript.

Tools Used : Android Studio, Visual Studio, App Scripts.

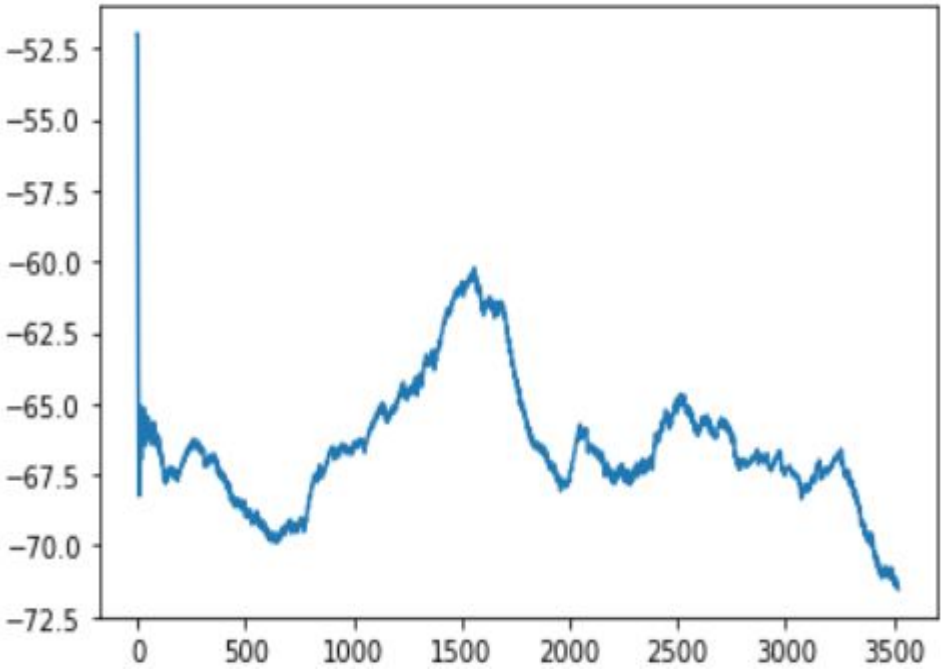
Application

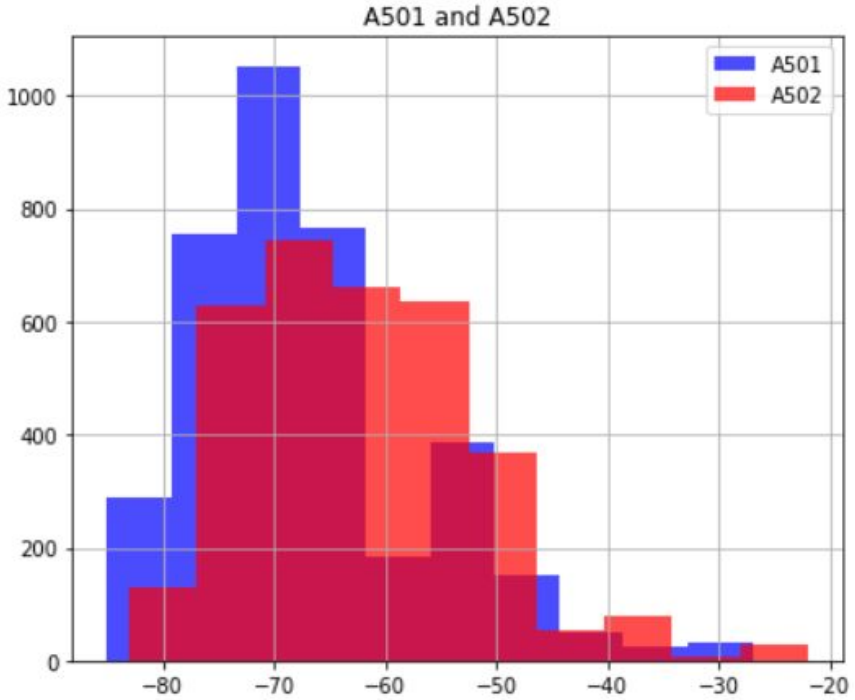
- We had created an Attendance Application which take input from Student.
- The User enter the details which is then processed and accordingly the correctness of the attendance is validated.
- We have used Wifi Access Points that are available across our Thapar University Campus to help localize the student for Fast, Hassle-free and secure attendance system.
- Finally the results can be seen by the Professor for their respective Courses.



Machine Learning

- After the data is collected from the students from the app in the form of google sheets, it undergoes preprocessing as well as feature engineering.
- An ML model has been applied on already available sample dataset (at github) containing information regarding various access points present in a warehouse.
- We have used XGBoost model on the dataset which is an open-source software library that provides a regularizing gradient boosting framework to deal with lots of erroneous data and noise.
  Moreover Overfitting will not happen easily.
- The data visualizations can be seen in the form of various graphs like signal strength and comparing the RSS of two different access points.





The above two graphs are very prominent from our analysis point of view as they tell us about the variation in signal strengths from an access point.

Also, RSSI of two access points i.e. A501 and A502 is compared.

Future Scope

- We would further be incorporating more ML models and carry out a thorough comparison amongst them.
- Then we would be integrating the final Ensemble Model into our Attendance Application.
- Furthermore we would creating a web interface , integrated with python that we would be performing the data preprocessing steps and would be emailing the final attendance sheets to the teachers email with the list of correctly and incorrectly marked attendance.

Thank You!